

SUPPLEMENTARY MATERIAL — Cross-Species CIS Study (v1.0, 2026-09-26)

Each section either contains the content inline or points to the frozen artifact that contains it (append-only; nothing regenerated).

S1. Dataset provenance

Zenodo 19796783 (AOMIC-ID1000 structural connectomes), CC-BY-4.0; 10 zips, 7,067,175,601 B, byte-exact vs API; CRC clean; 900 subjects $\times$ 7 atlases $\times$ 4 variants. Verification: `../00_Metadata/AOMIC_VERIFICATION_REPORT.md`; checksums: `../00_Metadata/CHECKSUMS_AOMIC.sha256`. Raw zips READ-FOREVER.

S2. QC rules

Pre-registered property-based QC (E02, CIS-blind): dimension mismatch, NaN, Inf, asymmetry $> 1e-9$, negative off-diagonals, isolated nodes, strength IQR, density IQR, multi-component (>1 non-trivial component). Cohort: 801/900 pass; 94 isolated-node + 5 strength-IQR flags; flagged subjects retained, never deleted. Artifacts: `03_BASELINE/qc_primary.csv`, `BASELINE_RESULTS.md`, `01_RAW_PROBES/raw_qc_report.csv`.

S3. CIS mathematical definition

$CIS(i) = (E(G) - E(G-i)) / E(G)$; $E(G) = (1/(N(N-1))) \sum_{i \neq j} 1/d(i,j)$ on the binary undirected thresholded graph; $E(G-i)$ on the induced $(N-1)$ -node subgraph ($(N-1)(N-2)$ normalization). Exact computation (scipy Dijkstra); reference-vs-naive validation $6.1e-16$ (`04_CIS/REFERENCE_IMPLEMENTATION_VALIDATION.md`); boolean-matmul fast path used only for null graphs, equivalence-gated at $1e-9$ (max dev $4e-16$; `05_NULLS/degree_preserving/FAST_IMPLEMENTATION_VALIDATION.md`). Implementation: `02_PREPROCESSING/cache_io.py` (single source of truth).

S4. Null-model algorithm

Undirected Maslov–Sneppen double-edge swaps: canonicalized edge list (deduplicated symmetric orientations), $n_swaps = 10 \times |E|$ attempts per null, rejecting self-loops, duplicates, degenerate picks; exact per-node degree verification per null (10,000/10,000 exact; 0 self-loops); rejection redraw (max 5 attempts). Adversarial regression suite (hub-selfloop/pendant, dense $p=0.6$, ring): `regression_tests_undirected.csv/.json`.

S5. Degree/strength matching

$\pm 10\%$ total degree, 1:1 greedy, no replacement; nearest-50 fallback flagged per pair. 36,846 degree-matched node pairs (top-decile CIS nodes vs matched controls). Residual defined as observed minus matched-control CIS; strength association reported alongside ($\rho(CIS, strength)$ median 0.486). Artifacts: `04_CIS/degree_control.py`, `degree_strength_control_summary.json`, `DEGREE_STRENGTH_CONTROL.md`.

S6. Threshold parameters

$k = \text{round}(c \cdot N(N-1)/2)$ largest |off-diagonal| weights; diagonal zeroed; binary output. Primary $c = 0.15$ ($k = 15,561$); ladder $\{0.10, 0.20, 0.25\}$. Rationale: raw densities 0.45–0.88 preclude unthresholded path-length statistics (STUDY_DESIGN §3.2). GE ladder at 4S456: 0.4911/0.5468/0.5877/ 0.6203.

S7. Statistical tests

- Battery B: per-subject z vs null distribution of top-50 median CIS; population two-sided sign test (binomial).
- Battery A: per-node empirical one-sided p with add-one correction ($p = (\#\{\text{nulls} \geq \text{obs}\} + 1)/(100 + 1) \rightarrow \text{Stouffer } z (\sum z / \sqrt{n_{\text{subjects}}}) \rightarrow \text{BH-FDR } q < .05$ (single stratum).
- Enrichment: control A = node-label shuffle (10,000 perms); control B = degree-matched peer pools resampled 10,000 $\times$; z and empirical p with add-one floor.
- Rank stability: 100 seeded half-splits, Spearman of mean-CIS ranks.

S8. FDR procedure

Benjamini–Hochberg over 456 node p -values (single stratum = primary configuration), $q = 0.05$; 43 survivors; censored nodes (all-null $\geq$ observed; 413 nodes) carry $p = 1$ and Stouffer $z = -\text{inf}$ by construction.

S9. Robustness configurations

9 conditions on frozen $n=150$ cohort: atlases $\{4S256, 4S156, \text{Brainnetome246Ext}, \text{AAL116}\}$, weights $\{\text{sift_invnodevol}, \text{radius2_count}\}$, costs $\{0.10, 0.20, 0.25\}$. All 9 conditions reproduce the primary architecture; max like-for-like top-50-mean deviation 0.00262 (atlas_AAL116; full precision 0.0026194, primary = 0.0012286 on the same $n=150$ cohort — RESOLUTION.md concordance). Note: no numeric tolerance was pre-registered for E06; the concordance is a descriptive comparison, not a pass/fail gate. Artifact: 06_ROBUSTNESS/, 09_TABLES/table_08_robustness.csv.

S10. Seeds

Subject samples: one RNG, seed 20260922, nested samples ($A \subset B$). Nulls: $100 + i$ per graph. Enrichment/rank-stability: 20260922. All recorded in null_manifest_battery_a.csv and script constants.

S11. Additional distributions

- Null δ/z distributions: battery_a/*.npz (100×456 null-CIS matrices per subject), null_results_battery_b.csv (per-subject null mean/sd).
- Per-subject CIS: 04_CIS/subject_cis/sub-XXXX.csv (456 rows each, 900 files).
- Baseline: 09_TABLES/table_baseline_network_properties.csv.

S12. Anatomical mapping

00_MANIFEST/manifests/atlas_4S456_system_labels.csv (456 nodes; Vis 61, SomMot 77, DorsAttn 46, SalVentAttn 47, Limbic 26, Cont 52, Default 91, Subcortical_Cerebellar 56). Human chokepoint catalogue: 07_CROSS_SCALE/human_chokepoint_catalogue.csv.

S13. Literature novelty matrix

10_LITERATURE/NOVELTY_MATRIX.md (+ Kudriavtsev gate record:
10_LITERATURE/KUDRIAVTSEV_2026_FULLTEXT_REVIEW.md).

S14. Reproducibility instructions

1. `git clone` the repository (heavy data excluded; raw zips from Zenodo).
2. Run `02_PREPROCESSING/preprocess_v2.py` (rebuilds `cache_parts/`, resume-safe) →
`03_BASELINE/baseline.py` → `04_CIS/cis_primary.py` → `05_NULLS/degree_preserving/null_models.py` --
battery `a|b` → `04_CIS/e05_statistics.py` → `06_ROBUSTNESS/robustness.py` →
`09_TABLES/build_tables.py` → `08_FIGURES/make_figures.py`.
3. Numeric audit without any recomputation: `python 10_REPORT/verify_final_numbers.py` (stdlib-only; 42 checks).
4. Environment: Python 3.13, numpy 2.4.2, scipy 1.18.0, pandas 2.3.3 (see `ANALYSIS_FREEZE.md` provenance-gap note).